Affine Richardson Geometry: Clusters, Links, and Catalan Combinatorics

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1 Introduction

My work focuses on positroid varieties and their Plücker patches. Positroid geometry grew out of total positivity: Postnikov described cells of the totally nonnegative Grassmannian using planar bicolored (plabic) networks [Pos06], and Knutson–Lam–Speyer realized their complex counterparts as projected Richardson varieties [KLS13]. Following Scott [Sco06], Serhiyenko–Sherman-Bennett–Williams and Galashin–Lam established positroid cluster structures described by these networks [SSBW19; GL23b]. Galashin–Lam related positroid cohomology to link homology, recovering rational q,t-Catalan polynomials for top cells with coprime k,n [GL24a]; their point counts yield lattice-path enumerations [GL24b]. Cluster constructions and rational Coxeter–Catalan enumerations now extend across finite Lie types [CGG+25b; GLTW24].

For Plücker patches, $\Pi^\circ_{f,I} = \Pi^\circ_f \cap \{\Delta_I \neq 0\}$, the positroid cluster structure is a starting point when Δ_I labels a plabic face, but an arbitrary minor may occur in no plabic seed [OPS15]. Snider identified these patches with affine Richardson varieties [Sni11]. The proposed research develops a periodic diagrammatic theory in affine type A: explicit plabic coordinates and links for these patches, with applications to Catalan bijections.

My preprint [1] introduces *affine Deograms*, periodic crossing-and-elbow diagrams encoding patch strata, which underpin my recursive bijection with rational Dyck paths [3]. My joint preprint with Matthew J. Tyler [2] connects Richardson torus actions to slice links. The three objectives build directly on these results:

1. **Affine cluster structures.** Realize affine plabic graphs as geometric seeds and prove that their cluster algebras equal the coordinate rings of affine braid Richardson varieties.
2. **Affine Richardson links.** Recover patch point counts from periodic link invariants, as a first step toward an affine cohomological correspondence.
3. **Catalan bijections and gradings.** Modify the first-elbow map to obtain Dyck-path and parking-function bijections beyond the Fuss case, and find intrinsic diagram statistics for q,t-Catalan polynomials.

2 Prior work and preliminary results

Write $\mathrm{Deo}^{\max}(k,n)$ for maximal Deograms of the top cell of $\mathrm{Gr}(k,n)$; these have the fewest elbows for fixed boundary data. Let $\mathrm{Dyck}(a,b)$ denote rational paths from $(0,0)$ to (b,a) above the diagonal.

Theorem 1 (Martinez [3])**.** *For coprime $0 < k < n$, there is an explicit recursive bijection*

$$\Phi_{k,n}\colon \mathrm{Deo}^{\max}(k,n) \xrightarrow{\sim} \mathrm{Dyck}(k,n-k).$$

[1]Numbered references indicate my work, including joint work.

In the Fuss–Catalan case I also obtain direct bijections. Scan the columns $1, 1-a, 1-2a, \ldots \pmod b$ cyclically; the scan position of the first elbow in row i gives the horizontal position of a north step labeled i. This produces a *parking function*, a Dyck path whose north-step labels $1, \ldots, a$ increase within each column, and erasing labels gives a Dyck path (Figure 1).

Theorem 2 (Martinez [4]). *Let $a, m \geq 1$ and $b = ma+1$. The first-elbow construction gives direct bijections*

$$\Psi_{a,b}\colon \mathrm{Deo}^{\max}_{a,b}(\mathrm{id}) \xrightarrow{\sim} \mathrm{Dyck}(a,b), \quad \widetilde{\Psi}_{a,b}\colon \bigsqcup_{u \in S_a} \mathrm{Deo}^{\max}_{a,b}(u) \xrightarrow{\sim} \mathrm{PF}(a,b).$$

Here $\mathrm{Deo}^{\max}_{a,b}(u)$ denotes maximal $a \times b$ rectangular Deograms with twist u, and $\mathrm{PF}(a,b)$ denotes rational parking functions.

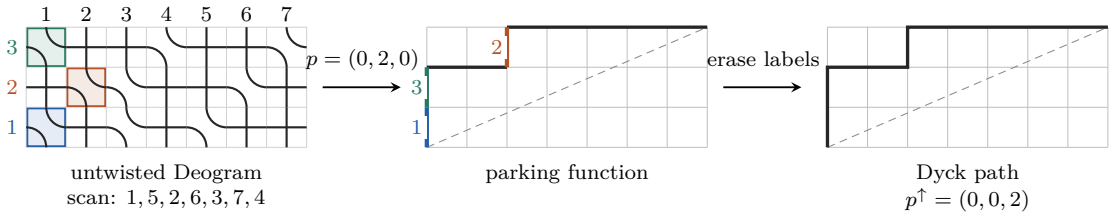


Figure 1: The direct Fuss–Catalan maps for $(a,b) = (3,7)$. Colored cells mark the first elbow in each row under the scan $1, 5, 2, 6, 3, 7, 4$. Their zero-based scan positions $p = (0, 2, 0)$ give the horizontal positions of north steps labeled $1, 2, 3$. Erasing labels gives the Dyck path.

An *affine braid Richardson variety* $X_{f,\beta}$ parametrizes affine flag chains starting at the standard flag, with steps prescribed by a positive word β and endpoint in the opposite Iwahori orbit indexed by f; common length-zero factors are suppressed. For reduced words, these varieties include positroid patches. Let $\delta(\beta)$ be the Demazure product. In work in progress with Moriah Elkin and Matthew J. Tyler, affine 3D plabic graphs assign a candidate exchange quiver $Q^{\mathrm{cand}}_{f,\beta}$, and we prove:

Theorem 3 (Elkin–Martinez–Tyler [5]). *For every positive affine word β and $f \leq \delta(\beta)$, the cluster algebra of $Q^{\mathrm{cand}}_{f,\beta}$ is locally acyclic.*

Thus the candidate cluster algebra equals its upper cluster algebra [Mul13]; identifying it with $\mathbb{C}[X_{f,\beta}]$ is the remaining geometric problem laid out in Objective 1 below.

For a finite positive word β, set $L_{u,\beta} = \widehat{\beta\beta(u)^{-1}}$, using the positive braid lift $\beta(u)$. A link is *smoothly slice* if it bounds disjoint smooth disks in the four-ball.

Theorem 4 (Martinez–Tyler [2]). *For $u \leq \delta(\beta)$ in finite type A, $X_{u,\beta}$ has a dense orbit under the standard diagonal torus if and only if $L_{u,\beta}$ is smoothly slice. For open positroid varieties, these conditions are also equivalent to the link being an unlink.*

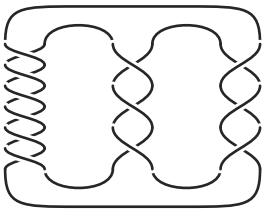


Figure 2: The nontrivial slice knot $12n_{582}$ corresponds to a Richardson variety isomorphic to $(\mathbb{C}^*)^7$, a single diagonal-torus orbit [2, §8.4].

3 Research Plans: Objectives, Methods, and Significance

Objective 1: Affine cluster structures

For a seed Σ, the cluster algebra $\mathcal{A}(\Sigma)$ is generated by the variables obtained through mutation, and the upper cluster algebra $\mathcal{U}(\Sigma)$ consists of functions Laurent in every seed. Finite Richardson and braid varieties admit explicit cluster constructions [Mén22; GLSBS22; CGG+25b]. Bao–Ye now identify coordinate rings of Kac–Moody Richardson varieties, including generalized Richardson varieties in twisted products, with upper cluster algebras [BY25]. The goal is an explicit affine plabic description and the equality $\mathcal{A} = \mathcal{U}$ for the geometric structure. For reduced words, identifying our locally acyclic graph seeds with theirs would prove their local acyclicity conjecture in affine type A [BY25, Conjecture 5.15].

Problem 1. Realize $Q^{\mathrm{cand}}_{f,\beta}$ by a seed $\Sigma_{f,\beta}$ of regular functions on $X_{f,\beta}$, compatible with Bao–Ye's upper cluster structure, so that

$$\mathbb{C}[X_{f,\beta}] = \mathcal{A}(\Sigma_{f,\beta}) = \mathcal{U}(\Sigma_{f,\beta}).$$

The test case in Figure 3 lies beyond ordinary plabic charts. For $f = [4, 5, 6, 9, 8, 7]$, the patch minor Δ_{246} cannot label a plabic source face: 246 and the boundary label 145 are not weakly separated, since their differences alternate cyclically [OPS15]. The periodic graph nevertheless produces a quiver with one mutable vertex. Writing x_j for the variable indexed by bridge b_j, its exchange is $x_8 x_8' = x_4 + x_5 x_7$.

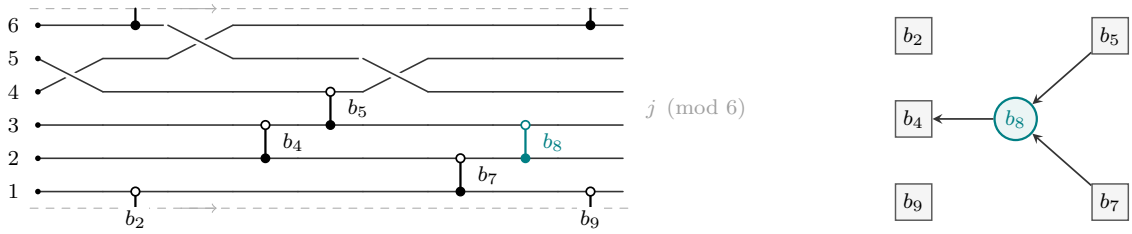


Figure 3: The affine graph for $\Pi^{\circ}_{f,246} \subset \mathrm{Gr}(3,6)$ and its candidate quiver. The dashed boundaries are identified. Teal marks bridge b_8 and the mutable vertex; frozen vertices are boxed. Arrows between frozen vertices are omitted.

The first target is to match reduced-word graph coordinates with Bao–Ye's generalized minors. The comparison must identify frozen units, including possible monomial factors, and respect mutation, including moves across the periodic boundary. Theorem 3 would then supply the coordinate-ring equality, and extending the comparison to flag chains addresses arbitrary positive words.

Objective 2: Affine Richardson links

The HOMFLY polynomial $P(L; a, z)$ is a two-variable isotopy invariant of oriented links, determined by $P(\text{unknot}) = 1$ and the *skein relation* below: reverse a crossing or smooth it, leaving the rest of the link unchanged [GL24a, §1.6].

$$aP\big(\ L_+\ \big) - a^{-1}P\big(\ L_-\ \big) = zP\big(\ L_0\ \big)$$

Galashin–Lam recover finite-type Richardson point counts by a normalized specialization of the highest-a term [GL24a]. For affine words, closing $\beta\beta(f)^{-1}$ gives a link in $T^2 \times [0,1]$, where $\beta(f)$ is a reduced braid lift [GN23]. The skein relation alone does not determine values on curves that wind around the torus; affine Hecke traces and monotone links suggest possible evaluations [GN23; GL23a].

Problem 2. Construct an invariant of periodic Richardson links whose normalized highest-a term satisfies

$$\mathcal{H}^{\mathrm{top}}_{f,\beta}(q) = \#X_{f,\beta}(\mathbb{F}_q) = \sum_D (q-1)^{e(D)} q^{b(D)},$$

and specializes to the finite-type HOMFLY formula.

The sum runs over Deodhar strata $(\mathbb{C}^*)^{e(D)} \times \mathbb{C}^{b(D)}$ [BD94]; $q-1$ and q count nonzero and arbitrary field elements. My Deogram recursions [1] provide tests on short affine words. The essential checks are braid relations and cut independence, with framing fixed; a subsequent homological refinement should recover cohomological gradings.

Objective 3: Catalan bijections and geometric gradings

For the top positroid cell and coprime k, n, the normalized point count at $q = 1$ is the rational Catalan number

$$\frac{1}{n}\binom{n}{k} = \#\,\mathrm{Dyck}(k, n-k),$$

connecting positroid geometry with rational noncrossing and nonnesting combinatorics [GL24b; GLTW24; ALW16]. Theorems 1 and 2 explain this ungraded count. Two problems remain: finding direct bijections beyond the Fuss case and explaining the two gradings in rational q,t-Catalan polynomials.

An exhaustive computation shows that the first-elbow map fails already for $(a,b) = (3,5)$: the 25 maximal rectangular Deograms over all twists yield only 24 distinct parking functions. Sorting the seven untwisted images likewise gives only five distinct Dyck paths. These collisions show that the rule must incorporate additional diagram data beyond the first-elbow positions.

Problem 3. Modify the first-elbow construction to obtain bijections with rational (a,b)-parking functions for arbitrary coprime $a \le b$, together with explicit inverses. On the untwisted family, the modified map should give a direct Dyck-path bijection.

The rational q,t-Catalan polynomial refines this count by weighting Dyck paths by their area and diagonal-inversion statistic dinv. Pulling these statistics back through Theorem 1 defines statistics on Deograms, but requires constructing the associated path. The goal is to read them directly from the Deogram itself.

Problem 4. For coprime $0 < k < n$, find intrinsic nonnegative integer statistics $A(D), B(D)$ on maximal Deograms D for the top cell of $\mathrm{Gr}(k,n)$ such that

$$\sum_D q^{A(D)} t^{B(D)} = \sum_{Q \in \mathrm{Dyck}(k,n-k)} q^{\mathrm{area}(Q)} t^{\mathrm{dinv}(Q)}.$$

In the Fuss case, Figure 1 illustrates how first-elbow positions recover area; an intrinsic formula for dinv remains to be found. An operation exchanging the two statistics would explain q,t-symmetry within the diagram model, and compatibility with the cohomological gradings of [GL24a] is the subsequent geometric question.

4 Career development

The Research Instructorship would support an independent program connecting combinatorics, geometry, and topology, built on my work on Deograms, Catalan bijections, and Richardson links. I aim to deepen my command of two areas: cluster comparison techniques, needed to prove regularity of diagrammatic coordinates, and link homology, needed to interpret combinatorial gradings geometrically. Working with my sponsoring scientist, Melissa Sherman-Bennett, would also help me identify problems in affine flag geometry beyond those proposed here.

I plan to organize a reading seminar on braid varieties and link homology, using concrete examples to connect their geometric, combinatorial, and topological descriptions, and to participate in the Davis algebra and topology seminars. In teaching and mentoring, I want to get better at designing problems, giving useful feedback, and guiding students from experimentation toward independent proofs.

5 Sponsoring scientist and host institution

UC Davis is an excellent institution for these research objectives: its faculty work across cluster algebras, braid varieties, link homology, and algebraic combinatorics, the areas this proposal combines. My sponsoring scientist, Melissa Sherman-Bennett, has developed cluster structures on Schubert varieties in the Grassmannian and 3D plabic graphs for braid varieties [SSBW19; GLSBS22], the foundation for Objective 1. Her work on toric Richardson varieties [GKSB26] also connects to my work on torus actions and links. Roger Casals and Eugene Gorsky's construction of braid variety cluster structures using weaves [CGG+25b], and its comparison with Deodhar constructions [CGG+25a], provide complementary tools for identifying the affine seeds. For Objective 2, Gorsky's work on affine Hecke traces [GN23] and Melissa Zhang's work on annular Khovanov homology [Zha18] would inform the link invariant and its cohomological refinement. For Objective 3, Monica Vazirani's work on affine permutations and rational parking functions [GMV16] and Erik Carlsson's work on the shuffle conjecture [CM18] provide models for extending the bijections and understanding their graded statistics.

6 Broader Impacts

Undergraduate research shaped my decision to pursue graduate study, so I have made mentoring undergraduates a priority at UCLA. Since 2023, I have mentored nine undergraduates through UCLA's Directed Reading Program and served on its organizing committee. My students have learned topics from the probabilistic method to polyhedral combinatorics, and most have presented at the symposium I co-organize. In spring 2026, Olha Shevchenko and I co-mentored three of these students on a project leading to a joint preprint on unimodality of q-Fibonomial coefficients [6].

I plan to keep mentoring and working with undergraduate and graduate students. Many projects coming out of my research are accessible to undergraduates and younger graduate students. For example, there are many natural maps on Deograms, and students could study how they relate to known maps on Fuss–Catalan objects such as the *zeta map*.